SOIL EROSION AND CONSERVATION · NOTE GC02

Gully Control by Runoff Management and Vegetation — Diversion, Sod Flumes, Sod Checks and Gully-Head Stabilisation

Course topics: SWCE4.5 L13, SWCE8.12 L8, SWCE8.15 L6

Controlling runoff from the drainage area

Controlling runoff is a major step in gully control. Order of preference: **retain** → **divert** → **convey**.

Approach	Methods and dimensions
Retention (store runoff in the drainage area)	Agronomic: crop management, contouring, strip cropping. Mechanical: bunding and terracing — contour bunds reduce or prevent runoff; graded terraces reduce the peak; contour furrows (small ditches or ridges across the slope) store water in proportion to their section and spacing; earth fills suit small and medium gullies on absorptive soils
Diversion (lead runoff away from the gully)	Terraces and diversion ditches — they remove runoff from the gully, reduce erosion and protect the outlet; used where slopes are too steep to terrace, on pasture and woodland. Filter strip ≥ 15 m wide to hold silt; channel capacity for a 10-year recurrence; diversion terrace ditches for watersheds of 3–3.5 ha; set the ditch back 3 × the height of the gully overfall from the gully head
Conveyance (pass runoff through the gully under control)	Vegetation, check dams and mechanical structures; temporary check dams protect until vegetation establishes

Vegetation in gully control

- Establish erosion-resistant cover — trees, shrubs, grasses or vines — to stabilise the gully, prevent erosion and provide income.
- **Spacing:** trees at most 2 m × 2 m; shrubs 1 m × 1 m or 1.2 m × 1.2 m. Fence the area 2 to 2.5 × the gully depth back from the gully edge.
- **Grasses:** *Cynodon dactylon*, *Dichanthium annulatum*, *Cenchrus ciliaris*, *Pennisetum purpureum*. **Trees:** *Prosopis juliflora*, *Acacia nilotica*, *Azadirachta indica*, *Albizzia lebbeck*, *Dalbergia sissoo*, *Eucalyptus* species.

Turning gullies into grassed waterways

Small and medium gullies with moderate runoff can be shaped to a broad, flat section that spreads water evenly, given a firm, fertilised seedbed, and seeded or sodded — permanent, economical, natural erosion control where good sod is available.

Sod flumes

- Control overfalls by giving the gully head a protected vegetative surface — for gully heads < 3 m high and drainage areas < 10 ha. They prevent waterfall erosion, protect the head and establish permanent vegetation cheaply and simply.

- Cut the overfall back far enough for vegetation to establish; the flume slope depends on soil, watershed size, overfall height and sod quality. Maximum depth of flow ≤ 30 cm.

Sod checks

Type	Use and dimensions
Sod strip check	Small gullies with flat grades and small to medium watersheds: sod strips laid across the channel, tops flush with or slightly below the bed. Minimum strip width 30 cm; 15 cm above the expected water level; strips 1.5–2 m apart. Easy, cheap, temporary
Sodded earth-fill check	Low sodded earth fills at regular intervals in small and medium gullies, each top level with the base of the next fill upstream. Side slopes at most 3 : 1 upstream and 4 : 1 downstream; height at most 45 cm; 15–30 cm above the maximum water level. A substitute for brush check dams
Shrub check	Shrubs planted in trenches across the flow line, 10–12.5 cm apart — for small drainage areas and mild gully slopes; slow the flow, induce silting and promote vegetation

Checking gully growth

- **Contour bunding** on slopes up to 6% — cross-section 0.9–1.3 m². In Gujarat ravines: 1.3 m² section, vertical interval 1.8–3.6 m, on slopes up to 12%.
- **Peripheral bunds** near gully heads control runoff and protect the head — cross-section 2.5 m², grade 0.1–0.2%.
- **Outlet structures** — pipe conduit, drop inlet, paved chute and stilling basin — dispose of runoff at non-erosive velocity.

Stabilising active gully heads

1. Slope the vertical head back to the natural angle of repose.
2. Fill the cave and face of the head up to two-thirds of its height.
3. Slope the remaining one-third.
4. Stabilise with vegetation (e.g. *Dichanthium annulatum*, *Cenchrus ciliaris*) and brush hurdles across the slope.

Pipe outlet (Fig. 15.11)	Dimension
Pipes	2 RCC pipes, 15 cm diameter
Opening	25 × 30 cm
Brick masonry box	0.61 × 0.61 m
Retaining wall length	3.66 m
Bund height (normal)	1.5 m
Flume length portion (approx.)	0.91 m
Bed width	120 cm
Foundation depth	80 cm

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. USDA Soil Conservation Service, Engineering Field Manual (gully control, sod structures). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The order of preference for controlling runoff in gully control is

(a) convey, divert, retain
(b) retain, divert, convey
(c) divert, retain, convey
(d) retain, convey, divert

Q2. A diversion ditch is set back from the gully head by about

(a) the gully depth
(b) 3 × overfall height
(c) 10 m
(d) 50 m

Q3. Minimum width of a filter strip with a diversion is

(a) 3 m
(b) 6 m
(c) 15 m
(d) 30 m

Q4. Sod flumes are used for gully heads less than

(a) 1 m high
(b) 3 m high
(c) 6 m high
(d) 9 m high

Q5. Maximum flow depth in a sod flume is

(a) 15 cm
(b) 30 cm
(c) 60 cm
(d) 1 m

Q6. Spacing of sod strip checks is

(a) 0.5 m
(b) 1.5–2 m
(c) 5 m
(d) 10 m

Q7. Maximum height of a sodded earth-fill check is

(a) 15 cm
(b) 30 cm
(c) 45 cm
(d) 90 cm

Q8. Peripheral bunds near gully heads have a cross-section of about

(a) 0.5 m²
(b) 1.3 m²
(c) 2.5 m²
(d) 5 m²

Q9. When stabilising a gully head, the cave is filled up to

(a) one-third
(b) half
(c) two-thirds
(d) full

Answers

Q1 — (b) retain, divert, convey
Q2 — (b) 3 × overfall height
Q3 — (c) 15 m
Q4 — (b) 3 m high — *and drainage area < 10 ha.*
Q5 — (b) 30 cm
Q6 — (b) 1.5–2 m
Q7 — (c) 45 cm
Q8 — (c) 2.5 m²
Q9 — (c) two-thirds — *of its height.*